StudyMate AI

Offline-First AI Study Assistant

for Snapdragon-Powered HP PCs

The Problem

- Students repeatedly upload personal notes to cloud AI tools
- Privacy and connectivity can be concerns
- Students need Q&A, summaries and practice questions
- A local-first workflow can keep study material on the PC

Our Solution

- Load PDF/TXT study material locally
- Ask questions and retrieve relevant passages
- Generate a quick extractive summary
- Generate simple practice MCQs
- No mandatory cloud API or API key

How It Works

Student notes → Local extraction → Local NLP → StudyMate AI →

Q&A / Summary / MCQs

Privacy-first design: the supplied study material remains on the device.

Snapdragon Optimization

Current prototype: lightweight local NLP baseline

Target deployment:

- Compact transformer/embedding model
- Qualcomm AI Hub evaluation and optimization
- Supported ONNX/Qualcomm execution path
- Target Snapdragon Hexagon NPU where supported

The submission clearly separates the working prototype from the planned Snapdragon optimization.

Technical Implementation

Python + Streamlit

PDF extraction with pypdf

Local TF-IDF retrieval with scikit-learn

Extractive summarization

Rule-based practice-question generation

Modular model layer for future replacement

Impact & Roadmap

Today: working offline-first prototype

Next: compact on-device model → Snapdragon benchmarking →

AI Hub optimization → desktop packaging

Goal: private, accessible AI study assistance on AI PCs

Thank You

StudyMate AI

A practical path from lightweight local NLP to Snapdragon-accelerated on-device AI.